1.5 Atomic structure and the periodic table

Students will be assessed on their ability to:

a recall the definitions of relative atomic mass, relative isotopic mass and relative molecular mass and understand that they are measured relative to 1/12th the mass of a ^{12}C atom

b demonstrate an understanding of the basic principles of a mass spectrometer and interpret data from a mass spectrometer to:

i deduce the isotopic composition of a sample of an element, e.g. polonium

ii deduce the relative atomic mass of an element

iii measure the relative molecular mass of a compound

c describe some uses of mass spectrometers, e.g. in radioactive dating, in space research, in sport to detect use of anabolic steroids, in the pharmaceutical industry to provide an identifier for compounds synthesised for possible identification as drugs

d recall and understand the definition of ionisation energies of gaseous atoms and that they are endothermic processes

e recall that ideas about electronic structure developed from:

i an understanding that successive ionisation energies provide evidence for the existence of quantum shells and the group to which the element belongs

ii an understanding that the first ionisation energy of successive elements provides evidence for electron sub-shells

f describe the shapes of electron density plots (or maps) for s and p orbitals

g predict the electronic structure and configuration of atoms of the elements from hydrogen to krypton inclusive using 1s …notation and electron-in-boxes notation (recall electrons populate orbits singly before pairing up)

h demonstrate an understanding that electronic structure determines the chemical properties of an element

i recall that the periodic table is divided into blocks, such as s, p and d

j represent data, in a graphical form, for elements 1 to 36 and use this to explain the meaning of the term *periodic property* k explain trends in the following properties of the element from periods 2 and 3 of the periodic table:

   i melting temperature of the elements based on given data using the structure and the bonding between the atoms or molecules of the element

   ii ionisation energy based on given data or recall of the shape of the plots of ionisation energy versus atomic number using ideas of electronic structure and the way that electron energy levels vary across the period.